

Does the Risk Model Still Matter Once the Investor’s Rules Are In?

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Blend-Ed

Background

Anyone investing money faces one decision: how much to put in each asset. Portfolio optimisation answers it by estimating each asset’s expected return and risk from past prices and solving for the best mix. In practice the estimates are noisy, and real investors add rules the textbook ignores: a limit on the number of holdings, a cash floor, sector caps, trading costs.

Real-world investing imposes constraints on portfolio optimisation. We ask how the choice of risk model changes the outcome once those constraints are fixed, and whether any model earns its complexity over the simplest alternative, equal weight.

What we ran. 5 investor rule-sets $\times$ 5 risk models (Markowitz, Ledoit–Wolf, robust, CVaR, equal weight), walk-forward over 2016–2025 with trading costs.

Limitations. One decade, one market (US large caps), no short selling or leverage, and a universe chosen with hindsight: all 40 stocks still trade today.

1. Experimental Setup

- **Universe:** 43 assets: 40 US large caps (four per GICS sector), a bond ETF, gold, cash.
- **Prices:** daily, 2008–2025. Tune on 2011–2015, test on 2016–2025.
- **Rule-sets:** five archetypes, values derived from Vanguard allocations and the 2022 Survey of Consumer Finances.

Archetype	CVaR limit start→end	max holdings	$w_{\max}$	cash floor	sector caps
Preserver	1.0→0.5%	20	8%	10%	IT 20%, Energy 10%
Steady Hand	1.5→0.7%	15	10%	5%	–
Builder	2.0→1.0%	12	12%	2%	–
Maverick	2.8→1.4%	8	20%	0%	–
Ethical	1.5→0.7%	15	10%	5%	excl. tobacco, oil, defence

2. The Portfolio Problem

$\mathbf{w} \in \mathbb{R}^n$ weights: fraction of the portfolio in each of n assets
 $z_a \in \{0, 1\}$ whether asset a is held
 $b_a, s_a \geq 0$ amount bought and sold, relative to the current weights $\mathbf{w}^{\text{prev}}$
 $\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}, r_s$ expected daily return, covariance, and day- s return vector, estimated from the last $|S|$ days
 $h = 21$, $\tau = 10$ bp expected holding period (days); cost per unit traded
 $\rho_{\max}(t)$ risk limit, tightening linearly as the horizon shrinks

$$\max_{\mathbf{w}, z, b, s} h \hat{\boldsymbol{\mu}}^\top \mathbf{w} - \tau \sum_a (b_a + s_a) \quad \text{s.t.} \quad \rho(\mathbf{w}) \leq \rho_{\max}(t)$$

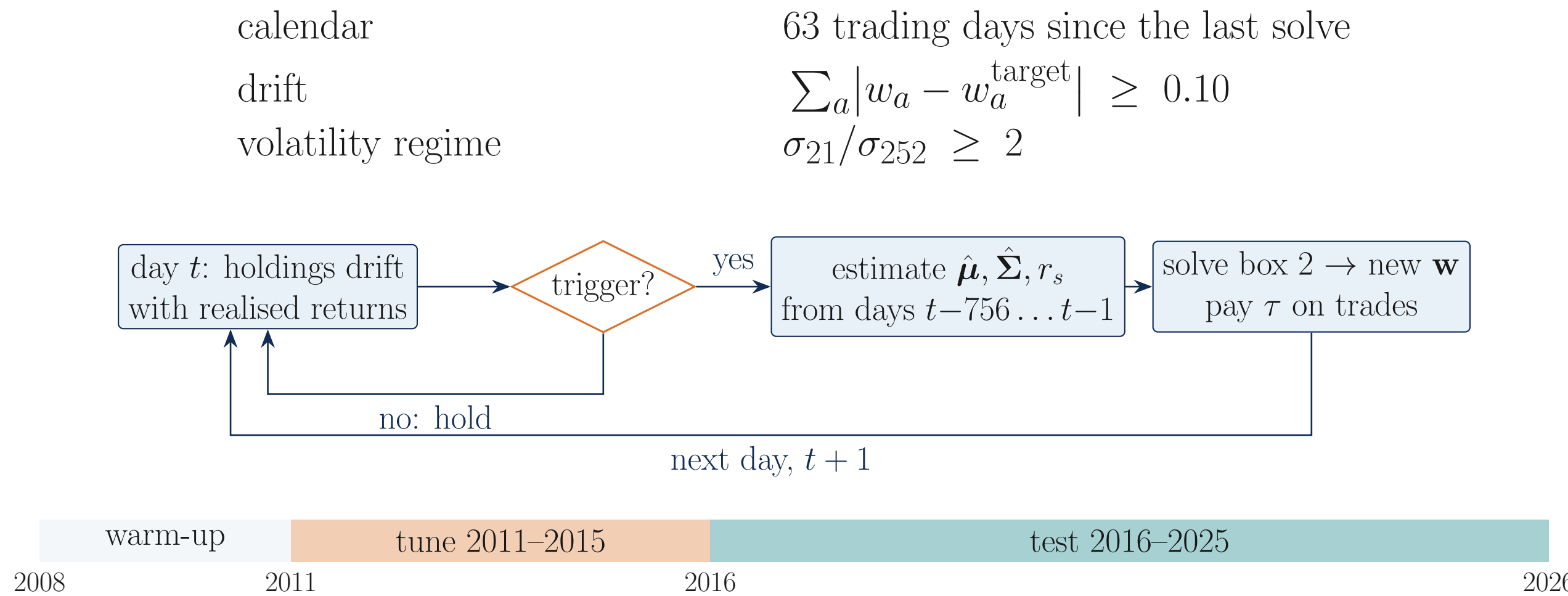
$\sum_a w_a = 1$ budget
 $w_a = w_a^{\text{prev}} + b_a - s_a$ trade balance
 $w_a^{\min} z_a \leq w_a \leq w_a^{\max} z_a$ minimum and maximum position
 $\sum_a \notin \text{cash } z_a \leq K$ cardinality: at most K names
 $\sum_{a \in \text{sector } k} w_a \leq c_k$ sector caps
 $w_{\text{cash}} \geq c_{\min}$ cash floor

3. Five Risk Models

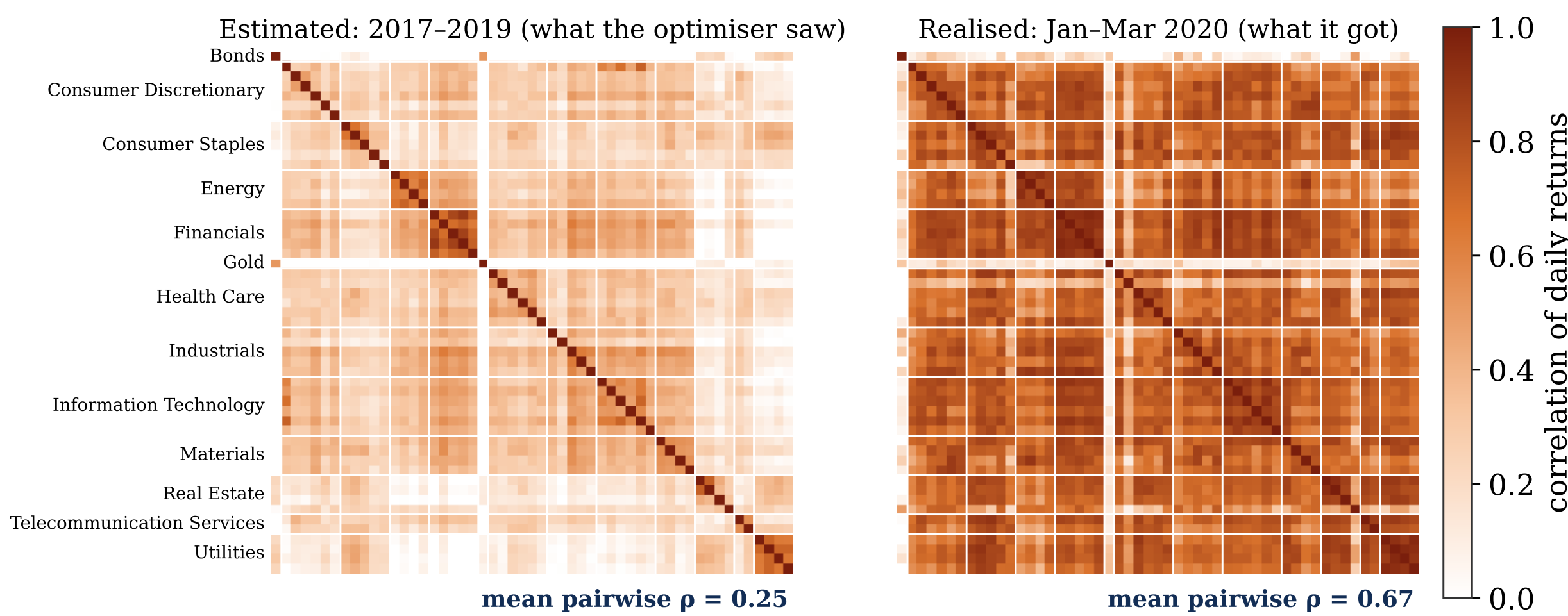
Markowitz $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\boldsymbol{\Sigma}} \mathbf{w}$
Markowitz + Ledoit–Wolf $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{\text{LW}} \mathbf{w}, \quad \hat{\boldsymbol{\Sigma}}_{\text{LW}} = (1 - \delta) \hat{\boldsymbol{\Sigma}} + \delta \bar{\sigma}^2 I$
Robust $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\boldsymbol{\Sigma}} \mathbf{w}, \quad \hat{\boldsymbol{\mu}}^\top \mathbf{w} \rightarrow \hat{\boldsymbol{\mu}}^\top \mathbf{w} - \kappa \sqrt{\mathbf{w}^\top \Omega \mathbf{w}}, \quad \Omega = \hat{\boldsymbol{\Sigma}}/|S|$
CVaR $\rho(\mathbf{w}) = \zeta + \frac{1}{(1 - \alpha)|S|} \sum_{s \in S} u_s, \quad \alpha = 0.95$
 $u_s \geq -r_s^\top \mathbf{w} - \zeta, \quad u_s \geq 0$
Equal weight $w_a = 1/N$

4. When the Portfolio Is Reweighted

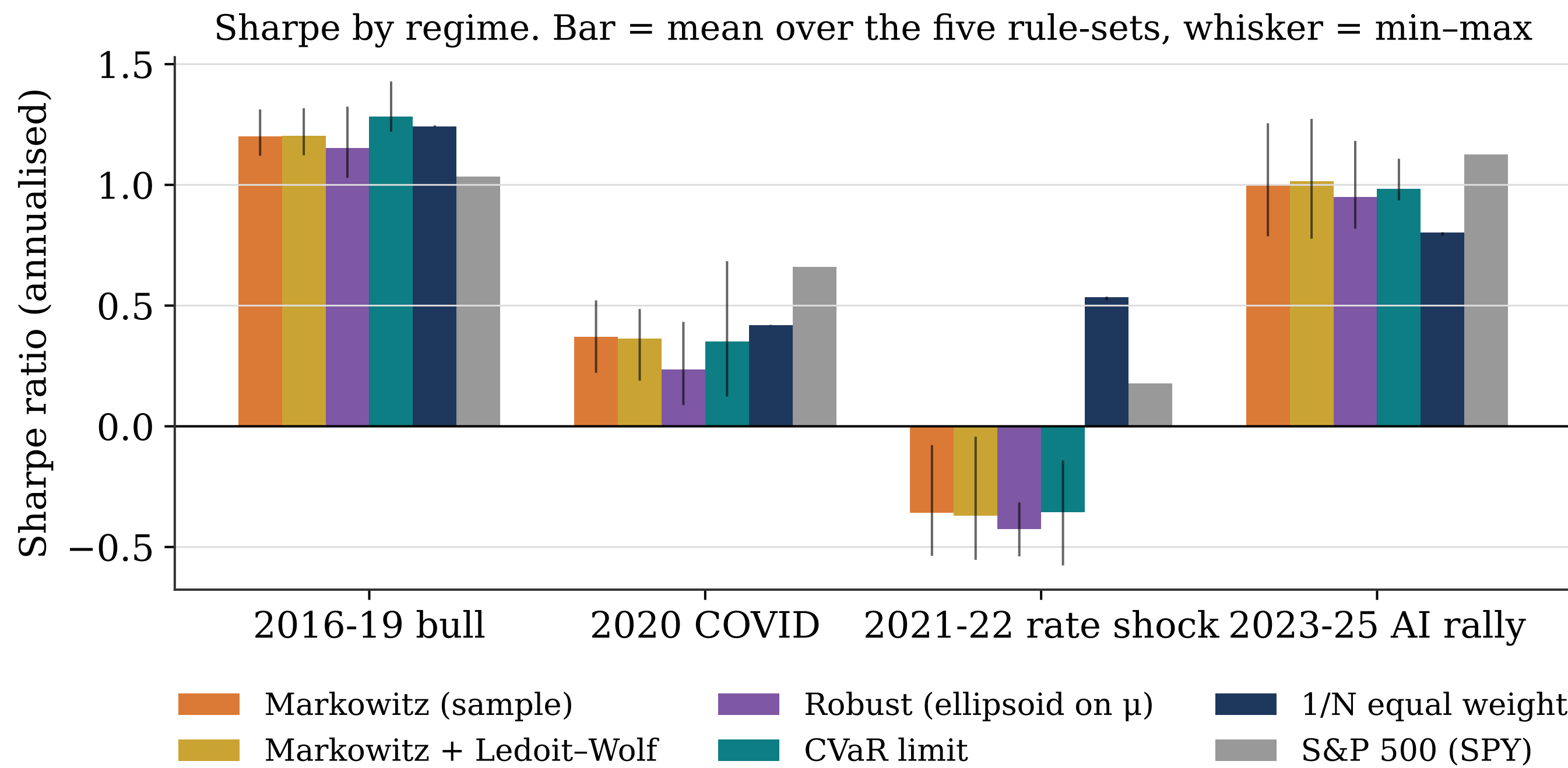
Weights are re-solved (box 2, from the current $\mathbf{w}^{\text{prev}}$) when any of the following holds:



5. Results I: When the Inputs Fail

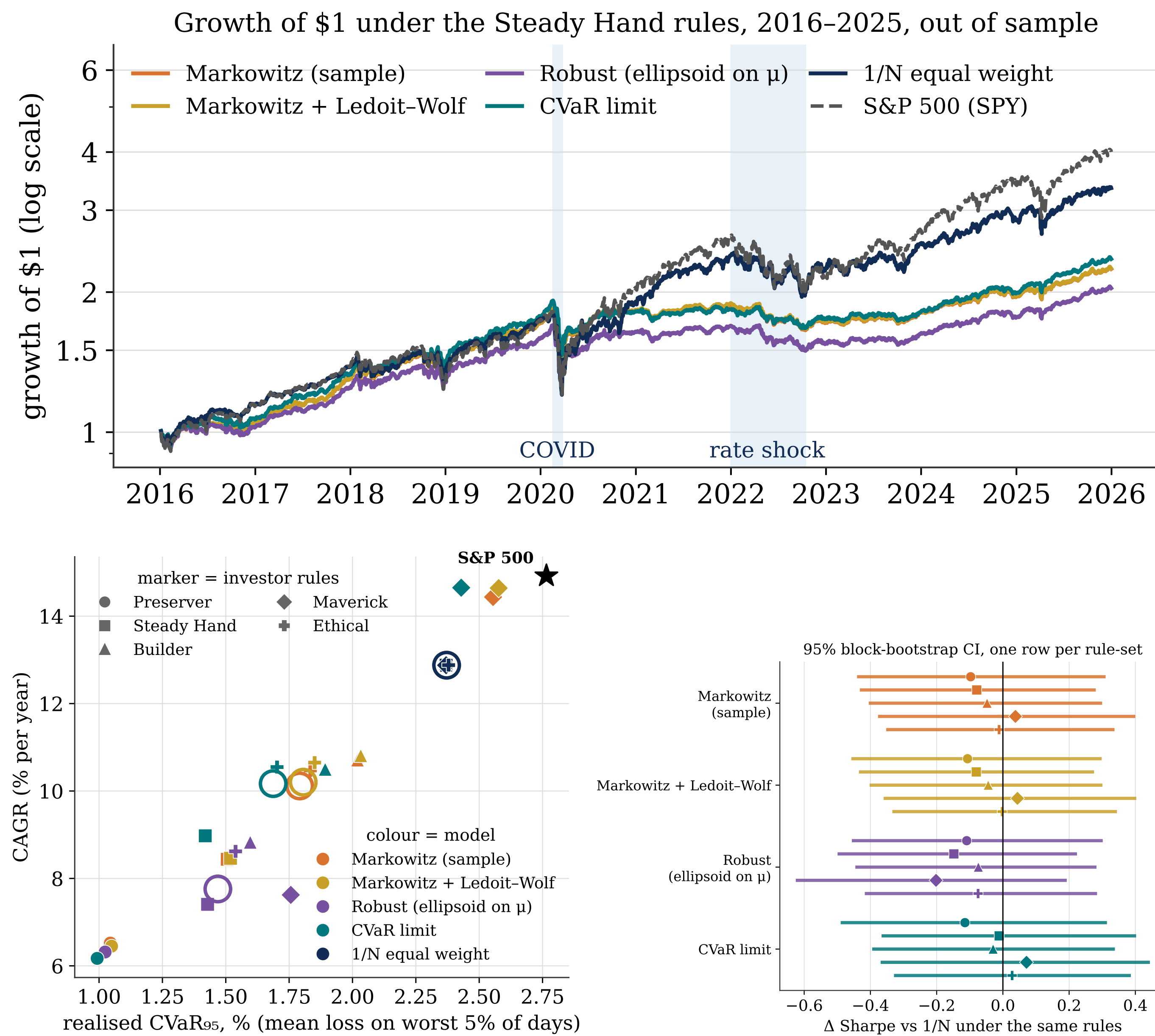


One event turned every pairwise estimate wrong at once: mean correlation $0.25 \rightarrow 0.67$. Every model in box 3 was fed the left-hand matrix and traded in the right-hand one.



2020: all models fell together, as the matrices predict. **2021–22:** every optimiser negative, 1/ N positive. A three-year lookback had loaded the optimisers onto the 2019–21 winners just before rates rose.

6. Results II: Rules vs Models



Left: rules decide risk, not models. Runs cluster by marker (rule-set), not by colour (model). **Right: no optimiser beats 1/ N .** Paired Sharpe difference under the same rules; 5000-draw block bootstrap; no interval clears zero.

Model	CAGR %	Sharpe [95% CI]	Max DD %	CVaR ₉₅ %
Markowitz (sample)	10.1	0.67 [0.14, 1.35]	-25	1.79
Markowitz + Ledoit-Wolf	10.2	0.67 [0.14, 1.35]	-25	1.81
Robust (ellipsoid on μ)	7.8	0.59 [0.03, 1.29]	-22	1.47
CVaR limit	10.2	0.70 [0.17, 1.40]	-25	1.69
1/ N equal weight	12.9	0.71 [0.19, 1.36]	-33	2.37
S&P 500 (SPY)	14.9	0.74 [0.20, 1.38]	-34	2.77

What the optimisers bought: 3–5 points less CAGR for ≈ 8 points less drawdown than 1/ N , at the same Sharpe. They delivered the risk target they were given.

7. Conclusions

- **The rules matter more than the model.** Four risk models within 0.04 Sharpe of each other and of 1/ N , inside bootstrap noise.
- **CVaR is the best optimiser and the cheapest** (linear, no covariance); still not better than 1/ N .
- **Robust is consistently worst:** guarding against the worst-case mean doubles up on rules that already limit concentration.
- **Nothing beats the S&P 500** in a mega-cap decade whose winners every rule-set caps. The optimisers bought drawdown, not return.

Interactive version: pick an investor, travel back to 2016 and watch each model trade.



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